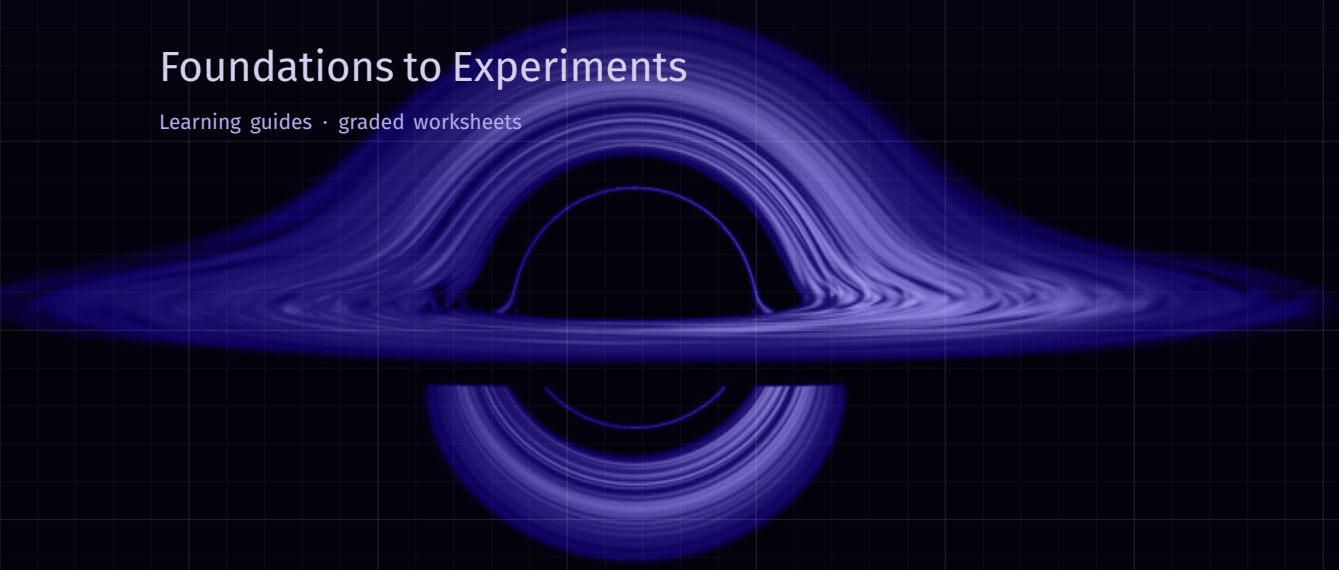
STUDENT WORKBOOK

From NLP to LLM

Student Workbook

Foundations to Experiments

Learning guides · graded worksheets

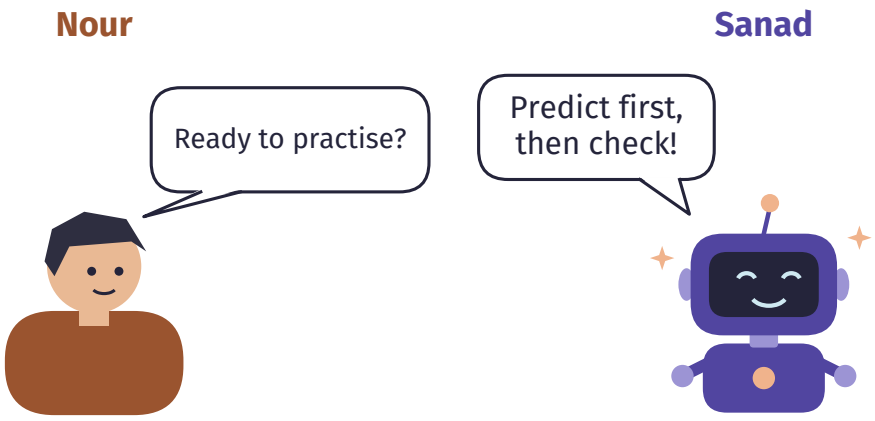


ONE PIPELINE · EVERY WORKSHEET



Mahmmoud A. Mahdi

FIRST EDITION · 2026



THIS WORKBOOK BELONGS TO

Student name _____

Student ID _____

Group / section _____

LEARNING GUIDE A / START HERE

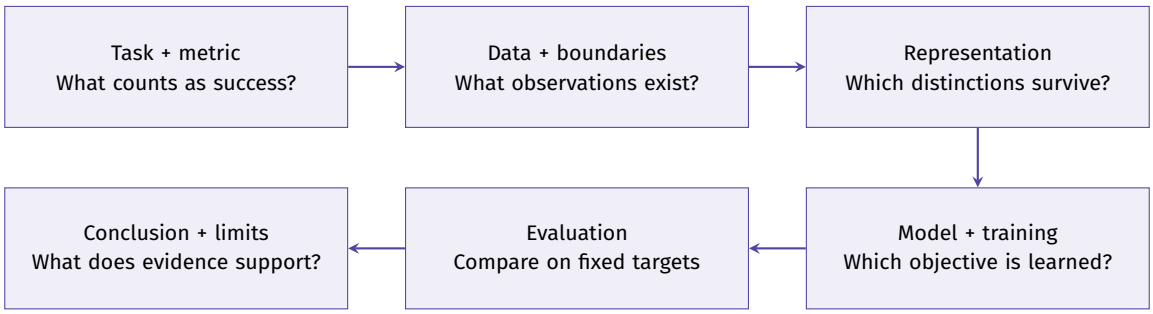
Connect the language-modelling concepts

Chapters 1–5 | Preparation before the worksheets



Goal. Explain how a language task becomes a representation, a probability model, and an experiment whose evidence you can interpret. Use the eight foundation chapters and the Chapter 9 assistant bridge, the worked examples, and the companion notebook. Predict small results before running code.

Figure 1. Follow the evidence chain



Read the arrows aloud: "For this task, using these data and units, I computed or trained this model, measured this result, and can support this conclusion." A fluent output alone does not complete that chain.

Your route through the packet

Sheets	Book chapter	Question	Carry forward
1–2, L1	1: foundations	What is the task and evidence?	Task contract and evaluation criteria.
3–4, L2	2: rules and patterns	What can a pattern show, and where does it fail?	Tested patterns, known failures.
5, L3	3: counts and distance	What does a corpus contain, and how far apart are two spellings?	Corpus counts, edit-distance tables.
6–8, L4	4: text units	What reaches the model?	Tokenizer, preprocessing policy, round-trip test.
9–11, L5	5: statistics	What do counts and features preserve?	Conditional and feature calculations, averaged metrics.
12–15, L6	6: neural lab	How does the network learn?	Shapes, derivatives, run evidence, decoding.
16–20, L7	7: experiments	Does the result generalise?	Split protocol, comparisons, final report.
21	8: assistants	Does the system help, not only predict?	Causal mask check, assistant evidence, two denominators.

Submission and working conventions

Write your name and date on each sheet. Keep calculations, configurations, source revisions, commands, and outputs together. Attach extra tables or logs when a page cannot hold the full result. Number attachments and refer to them from the worksheet.

The learning guides contain worked preparation; the numbered sheets provide space for your own answers. Sheets 1–15 develop chapter practice for Chapters 1–7, sheets 16–20 organise the Chapter 8 assignment tasks A1–A6, and sheet 21 evaluates an assistant system for Chapter 9. Each Sanad project level in the Lab Manual has a project log, placed after its chapter's sheets (L1 after Chapter 1, L2 after Chapter 2, L3 after Chapter 3). Tasks marked CHALLENGE belong to the challenge route: they are optional, and a core submission is complete without them. The Arabic extension A6 is optional. No course marks are assigned by this packet.

Readiness check

Explain the difference between training loss and held-out evidence. Name one input distinction your task must

preserve.

LEARNING GUIDE B / FOLLOW ONE EXAMPLE

From text to prediction

Worked preparation | Separate two-sentence practice corpus

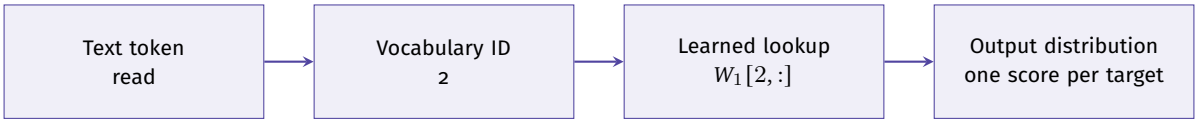


Practice corpus: “we read books” and “we read notes”. Lowercase, split on spaces, keep sentence boundaries, and use no start/end tokens. This is a preparation example, not the six-sentence workbook corpus.

Make every representation explicit

Stage	Worked result
Vocabulary	Alphabetical order: books, notes, read, we. IDs: 0, 1, 2, 3.
Pairs	(we, read), (read, books), (we, read), (read, notes). No books-to-we transition crosses the boundary.
Input / target IDs	$(3, 2)$, $(2, 0)$, $(3, 2)$, $(2, 1)$. There are $N = 4$ examples and $V = 4$ words.
One-hot contract	For (we, read), input $(0, 0, 0, 1)$ and target $(0, 0, 1, 0)$ under this mapping.
Count prediction	Two outgoing read transitions: one to books, one to notes. Each conditional probability is $1/2$.

Figure 2. Identity is not a probability



The integer 2 chooses a row. It is not a semantic magnitude or a probability. Changing vocabulary order requires corresponding changes to input and output indexing.

A complete conditional calculation

For context read, $C_{\text{out}} = 2$. Add-one smoothing over the four words gives books and notes $2/6$ each; read and we get $1/6$ each. These sum to one. If the support included a new end marker, the denominator and the training transitions would change.

This unpadded example learns next-word transitions. It does not learn how to begin or end a sentence. The final words books and notes have no outgoing evidence.

Try it on a new context

Calculate the unsmoothed and add-one distributions after we. Show all four target probabilities and check the sums. Explain why the two methods differ.

Transfer: repeat the reasoning on the supplied six-sentence corpus on sheets 9 and 12. Keep the two corpus names distinct.

WORKSHEET 1 / CHAPTER 1

Define the language task

Chapter 1 E1 | Local objectives O1, O2 and O6



Name: _____ Date: _____

> **Shared case.** A university help desk receives English, Arabic, and mixed-language requests about course access, assessment, and registration. A request may be ambiguous or lack a documented procedure. Design a task whose outputs and evidence can be checked.

1. Write a task contract

Use Chapter 1's worked help-desk specification as preparation. Adapt its routing and clarification policy to your campus; distinguish routing, urgency, and procedural advice. Specify available context, reference labels, multi-intent handling, important error, metric, and fallback. Label any target as a proposal needing agreement.

Contract item	Your specification
Task / input / context	
Labels / clarification	
Reference / varieties	
Error / metric / target	
Fallback / owner	

2. Analyse ambiguity before choosing a model

Write one ambiguous request, two plausible interpretations, and the contextual evidence or clarification needed to distinguish them. Identify the ambiguity type.

3. State Arabic requirements

Name the varieties, mixed-script cases, and morphological or diacritic distinctions your evaluation must include. State what information should be preserved.

Carry forward: use this contract on sheets 2 and 20. A routing score does not verify generated instructions.

WORKSHEET 2 / CHAPTER 1

Choose meaningful evidence

Chapter 1 E1, E3 and E4 | Local objectives O3–O5



Name: _____ Date: _____

1. Calculate urgent-ticket metrics

A constructed evaluation has $TP = 80$, $FN = 20$, $FP = 60$, and $TN = 840$. Compute each metric, showing its denominator. Compare with the book's 60/40/40/860 example.

Metric	Calculation / result	What this tells the help desk
Accuracy		
Precision		
Recall		
Urgent-class F_1		

Same accuracy, different consequences. Which error does each model make more often, and which matters more to the help desk?

2. Compose modalities and locate an error

Use a spoken request about a screenshot. Draw the stages, label each input/output modality, and identify an error that can cross an interface. Compare a pipeline with a direct multimodal model.

Stage measure, expected end-to-end result, and comparison evidence:

3. Explain the engineering choice

Compare a rule-based or count-based baseline with a reusable neural model for your contract. State one advantage, one limitation, and the evidence needed to choose between them.

PROJECT LOG L1 / CHAPTER 1

The yardstick

Lab Manual, project level L1 | Dev split, 20 tickets



Name: _____ Date: _____

> **Replace, measure, then decide.** Run `python -m sanad report` before and after each change, on the same dev split and shared parts. Keep the old version when the evidence does not support the new one.

1. Router baseline: vo_majority

Record	Old: (none)	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

2. Search baseline: vo_exact

Record	Old: (none)	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

Decision

Which versions does your `sanad.toml` select now, and which result would make you revert one of them?

REVIEW / CHAPTER 1

NLP Foundations and Applications

Answer space for every item in Section 1.7, Review and practice



Name: _____ Date: _____

Quick quiz

Circle one letter for each question and give your reason in a few words.

1	A	B	C	D	Reason _____
2	A	B	C	D	Reason _____
3	A	B	C	D	Reason _____
4	A	B	C	D	Reason _____
5	A	B	C	D	Reason _____
6	A	B	C	D	Reason _____
7	A	B	C	D	Reason _____
8	A	B	C	D	Reason _____
9	A	B	C	D	Reason _____

Check your understanding

Q1 How do NLU and NLG differ in the course-help-desk example? Name an output and a possible failure for each.

Q2 What changes when “big” becomes “small” in the trophy–suitcase example? Name that kind of ambiguity, then give one pragmatic and one morphological example from the chapter.

Q3 Which engineering decisions remain after replacing rules with a pre-trained model?

Q4 Why can accuracy hide failure on urgent tickets, and which baseline should be reported beside it?

Q5 A voice help desk chains recognition, routing, and speech synthesis. Why measure each stage *and* the end-to-end result?

Q6 Why should an Arabic task specify dialect and diacritic policy? Name a help-desk input that an edited Modern Standard Arabic evaluation could miss.

Work it through

E1 Write a one-paragraph task contract for a course help desk adapted to your campus: input and available context, routing output, multi-intent and clarification policy, reference labels, language varieties, important error, metric, proposed target, and fallback. Include one ambiguous example and the question needed to clarify it. Justify a rule-based baseline or reusable model as a first candidate; do not assume that a newer model is better.

E2 Measure a router yourself. Ten held-out tickets have these reference (Ref) and predicted (Pred) intents, where A = access, S = assessment, R = registration, and O = other:

Ticket	1	2	3	4	5	6	7	8	9	10
Ref	A	A	A	S	S	R	R	O	A	S
Pred	A	A	A	S	A	R	S	O	A	S

The most frequent label in the training tickets was A. Mark each ticket right or wrong, then compute accuracy and the majority baseline on these tickets. Build the 4×4 confusion matrix, and calculate recall for each class.

E3 Recalculate precision, recall, F1, and accuracy for 80 true positives, 20 false negatives, 60 false positives, and 840 true negatives. Compare with Example 1.2. Explain how equal accuracy can conceal different missed-ticket and false-alarm costs.

E4 Design a help-desk scenario that uses text, speech, and an image, such as a spoken request about a screenshot. Name the stages and their input/output modalities, an error that can cross an interface, and one end-to-end test with an expected result. Compare a pipeline with a direct multimodal model and name the evidence needed to choose.

Read the picture

V1 Redraw Figure 1.7 for a classifier that still finds 60 urgent tickets but flags only 20 ordinary ones. Which measures change?

V2 In Figure 1.9, where would all 1000 tickets go for the majority baseline that predicts “not urgent” every time?

WORKSHEET 3 / CHAPTER 2

Predict, then score a pattern

Chapter 2, Section 2.1 and Figure 2.6; Lab 1 | Local objectives O1, O3 and O5



Name: _____ Date: _____

1. search or fullmatch?

Predict in pencil, then check with Python and correct in pen. Write what `re.search` returns (or "none"), and whether `re.fullmatch` succeeds.

Pattern	Input	search returns	fullmatch?
`\d{3}`	`room 2045`		
`[A-Z]{2,4}\s?\d{3}`	`CS 101`		
`[A-Z]{2,4}\s?\d{3}`	`cs101`		
`EGP ?\d+`	`EGP 75`		
`\bexam\b`	`exams`		

2. Score spans exactly

The pattern is `\b[A-Z]{2,4}\s?[0-9]{3}\b`. For each message, write the spans it returns, then mark each true code and each returned span as TP, FP, or FN (Figure 2.6).

Message	True codes	Returned spans	TP / FP / FN
Moved CS 210 to HALL 105	CS 210		
Missing grade in cs210	cs210		
NLP 301 and MATH 205, the 205 written in Arabic-Indic digits	NLP 301, MATH 205		
Section CS101-B is full	CS101		

Totals, precision, and recall (show what you divided by): _____

3. Trace the clitic rule N8

N8 strips an optional و/ف, then an optional ل/ك/ب, then an optional ال, but only if at least three letters remain.

Word	N8 output	Correct?	Property of Arabic involved
والطلاب			
وصل			
للطلاب			
فكتب			

WORKSHEET 4 / CHAPTER 2

Predict patterns, then reflect

Chapter 2, Figures 2.2, 2.7–2.10; Section 2.3; Lab 2 | Local objectives O1, O2 and O4



Name: _____ Date: _____

Predict each result before you test it with Python's `re` module.

1. Predict what a pattern accepts

Mark √ if the pattern matches the *whole* string and × if not. Circle any result that is wrong for the task.

Pattern	الكتاب	الكتب	كتب
كتب	كتاب + ال			
(كتب | كتاب:?) + ال			

Replace `(\w)\1{2,}` with `\1`	*gamiiiil*	*gameel*	*hhhhhhh* (laughing)
Output			
Keep the rule, or exempt this case?			

2. Predict what price pattern V3 returns

V3 is "`EGP`, then a number" or "a number, then جنيه or ج.م" (Listing 2.2). Write what it returns, and whether that is right.

Message	**V3 returns**	**Right? If not, why not?**
`EGP 1,250`		
75 جنيه		
fee 20.00 `EGP`		
`EGP` 150.50 and 20 جنيه		

Lookaround	**Your answer**
A lookbehind that returns the digits after رقم	
Why `(?<=رقم\s*)` fails in Python, and the fix	

3. Trace one ELIZA turn

Input: أنا قلق على مستقبلي ("I am worried about my future"). Write each stage in Arabic.

Stage	**Result / mechanism**
Keyword and rank	
Captured text	
Reflected text → reply	
An input where reflection fails, and why	

PROJECT LOG L2 / CHAPTER 2

Rules, slots, and a clarifier

Lab Manual, project level L2 | Dev split, 20 tickets



Name: _____ Date: _____

> **Replace, measure, then decide.** Run `python -m sanad report` before and after each change, on the same dev split and shared parts. Keep the old version when the evidence does not support the new one.

1. Router: v0_majority → v1_rules

Record	Old: v0_majority	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

2. Shared normaliser: v0_identity → v1_policies

Record	Old: v0_identity	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

Decision

Which versions does your `sanad.toml` select now, and which result would make you revert one of them?

REVIEW / CHAPTER 2

Rules and Patterns

Answer space for every item in Section 2.4, Review and practice



Name: _____ Date: _____

Quick quiz

Circle one letter for each question and give your reason in a few words. Items 12 are challenge route CHALLENGE.

1 (A) (B) (C) (D) Reason _____

2 (A) (B) (C) (D) Reason _____

3 (A) (B) (C) (D) Reason _____

4 (A) (B) (C) (D) Reason _____

5 (A) (B) (C) (D) Reason _____

6 (A) (B) (C) (D) Reason _____

7 (A) (B) (C) (D) Reason _____

8 (A) (B) (C) (D) Reason _____

9 (A) (B) (C) (D) Reason _____

10 (A) (B) (C) (D) Reason _____

11 (A) (B) (C) (D) Reason _____

12 (A) (B) (C) (D) CHALLENGE Reason _____

Check your understanding

Q1 In Python, `\w+` breaks a fully vocalised Arabic word into pieces. Why? Give two ways to fix it.

Q2 Name one task where alef normalisation (N3) helps, and one task where it does harm.

Q3 Policy N8 will not split فكتب (*fa-kataba*). Explain why, using the word “greedy”.

Q4 Look at the two shaded rows in Figure 2.15. Which ELIZA mechanism causes each failure?

Q5 A set of routing rules is right on almost every message it was written from, but on far fewer new messages. Why? Which messages should you write rules from, and which should you score them on?

Q6 Policy N3 and the `tolerant` search pattern of W2 both treat الإختبار and الاختبار as the same word. What does each one change, and when would you choose the pattern instead of normalising?

Work it through

E1 Write a pattern that finds dates written as day/month, such as 15/10. Each part has one or two digits, in ASCII or Arabic-Indic. Give two inputs it must accept, two it must reject, and one wrong input it still accepts.

E2 A pattern returns 30 spans; 24 are correct. The messages contain 32 true spans. Compute precision and recall. Then say what could happen to each one if you change the pattern to accept lower case too.

E3 V3 of the price pattern in Example 2.3 returns only `1` from `EGP 1,250`. Write V4 so it returns `1,250`. Name one new false positive V4 might let in, and one more input it should accept (the Arabic thousands separator is U+066C).

E4 **CHALLENGE** Extend the time pattern of W5 so the full words صباحًا and مساءً, with or without the tanween mark, are read as a.m. and p.m. Test it on ٧:٠٠ مساءً, on ١٠:٣٠ ص, and on ٩:٠٠ مدرج ٥. Which earlier policy would let you write the words without the mark?

E5 CHALLENGE The question detector of W6 misses فين القاعة (“the hall, where?”). Add a third signal: a question word at the *end* of the message. Give one statement that your new rule wrongly calls a question.

E6 Use a lookbehind to return only the amount from the message رقم 4589 والمبلغ 300 (“number 4589 and the amount 300”). Then write one `re.sub` call that turns `CS 120` and `CS120` into `<CS120>`, using two capturing groups. Why can the lookbehind not allow any number of spaces?

Read the picture

V1 In Figure 2.8, a new version removes a false positive and changes nothing else. Which of P and R must rise, and which cannot change?

V2 In Figure 2.14, what would the reply look like without the reflection step, and why?

WORKSHEET 5 / CHAPTER 3

Count, then fill a table

Chapter 3, Sections 3.1–3.2; Lab 3 part C preview | Local objectives O1 and O3



Name: _____ Date: _____

1. Count tokens and types

Message: الامتحان غدا والامتحان صعب (“the exam is tomorrow, and the exam is hard”).

Tokenization policy	*N*	*V*	The types
Split on spaces only			
Also split و off as its own token			

The Lab 3 corpus has $N = 185$ and $V = 105$. Its type–token ratio is _____. *Preparation only:* you compute these counts yourself in Lab 3, part C.

2. Fill a Levenshtein table

Warm-up. In Example 3.3 and Figure 3.8, Levenshtein charges 2 for the typo *exma* → *exam*. Which move is it missing, and what would the distance be with it? _____

Source درس (*dars*, “lesson”), target مدرسة (*madrasa*, “school”). All costs are 1. Use the method of Figure 3.7. As in the book, the source runs down the rows and the target along the columns, with letters in reading order. The first row and column are filled in; fill the rest, row by row, then shade one best path.

	#	م	د	ر	س	ة
#	0	1	2	3	4	5
د	1					
ر	2					
س	3					

Distance, and the operations on your path: _____

WORKSHEET 5 / CHAPTER 3, CONTINUED

Read the policy behind a number

Chapter 3, Example 3.1, Figures 3.4 and 3.12 | Local objectives O1–O3



3. Predict what a counting command keeps

Message: `The exam's at 10. the EXAM room is 2B.` Write the lines that `tr -cs 'A-Za-z' '\n'` produces, before you run it.

Item	Your answer
The tokens, in order	
N and *V*	
Which tokens are fragments, and what was lost?	
V if the text is lower-cased first	

4. Word form, lemma, root

Fill in the dictionary word (lemma) and the root of each form. Then count the lemmas and the roots.

Form	Lemma	Root
يَكْتُبُ (“he writes”)		
كَتَبُوا (“they wrote”)		
كُتُب (“books”)		
كَاتِب (“writer”)		
مَكْتَبَة (“library”)		
مَكَاتِب (“offices”)		

Lemmas: _____ Roots: _____ Which form looks like a verb without its vowel marks? _____

5. Normalise, measure, or check?

For each case, write **N** (normalise), **D** (edit distance), or **C** (a separate check is needed), and why.

Case	N/D/C	Why
أمتحان and امتحان in a search box		
Typos that users invent, such as امتحن		
مدرسة and مدرسه		
Are “IBM” and “IBM Inc.” the same company?		
Arabizi *3ayez*, *3ayz*, *3aiz*		

Now predict Sanad’s answer for each query, with the lexicon and threshold of Example 3.5 (normalise with N3–N4 first). Then say whether the suggestion helps.

Query	Suggestion	Edits	Helps?	What would fix it?
نتيجة				
جداول (“timetables”)				
غائب (“absent”)				
تسجبل				
مكتبتي (“my library”)				

6. Write a data statement

Sanad’s team saves 300 messages from the university help-desk chat during the January exam week. Fill in the statement (Section 3.1.1) before anyone counts a word.

Part	Your statement
Purpose, source, time, and how messages were chosen	
Varieties and writers: what you know, and what you must not guess	
Consent and personal data (names, IDs, phone numbers)	
Cleaning and tokenization rules, and how duplicates are handled	
One claim its counts cannot support, and why	

WORKSHEET 5 / CHALLENGE

Trace an alignment and a word error rate



Chapter 3, Sections 3.2.1–3.2.5 CHALLENGE | Local objective O4

Name: _____ Date: _____

7. CHALLENGE Backtrace a filled table

Source كتب (*katab*, "he wrote"), target مكتوب (*maktub*, "written"), letters in reading order. All costs are 1. The table is filled in. Start at the bottom-right cell and walk back to the top-left one, testing the diagonal first. Shade your path.

	#	م	ك	ت	و	ب
#	0	1	2	3	4	5
ك	1	1	1	2	3	4
ت	2	2	2	1	2	3
ب	3	3	3	2	2	2

Item	Your answer
Operations, in reading order (M, S, I, D)	
Distance	
With substitution cost 2, does the distance change? Why?	

8. CHALLENGE Word error rate

Follow the four steps of Section 3.2.5. Use the Egyptian pair in Figure 3.15.

Item	Your answer
S, *D*, *I*, and *N*	
WER before merging عاوز/عايز	
WER after merging them, and why it changed	
Add the pair ايوه → لا ("yes" heard as "no"). The pooled WER of both pairs, before merging; the mean of the two sentence WERs, and why you should not report it	

9. CHALLENGE Add a swap move

Nour types مدسرة for مدرسة: two neighbouring letters swapped.

Item	Your answer
Levenshtein distance, and with a swap move at cost 1 (Figure 3.8)	
Which earlier cell must a swap step read, as $D[\cdot][\cdot]$?	

PROJECT LOG L3 / CHAPTER 3

Corpus counts and fuzzy search

Lab Manual, project level L3 | Dev split, 20 tickets



Name: _____ Date: _____

> **Replace, measure, then decide.** Run `python -m sanad report` before and after each change, on the same dev split and shared parts. Keep the old version when the evidence does not support the new one.

1. Describe the training tickets (`python -m sanad corpus`)

Variety	Tokens N	Types V	Hapax	V/N
en				
msa				
egy				
arabizi				
all, and the shared parts used				

2. Search: vo_exact → v1_fuzzy

Record	Old: v0_exact	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

Decision

Which versions does your `sanad.toml` select now, and which result would make you revert one of them?

REVIEW / CHAPTER 3

Counting and Comparing Words

Answer space for every item in Section 3.3, Review and practice



Name: _____ Date: _____

Quick quiz

Circle one letter for each question and give your reason in a few words. Items 9, 10, 11 are challenge route CHALLENGE.

1 A B C D Reason _____

2 A B C D Reason _____

3 A B C D Reason _____

4 A B C D Reason _____

5 A B C D Reason _____

6 A B C D Reason _____

7 A B C D Reason _____

8 A B C D Reason _____

9 A B C D CHALLENGE Reason _____

10 A B C D CHALLENGE Reason _____

11 A B C D CHALLENGE Reason _____

Check your understanding

Q1 Two reports give different vocabulary sizes for the same messages. Neither is wrong. How can that be, and what must each report state?

Q2 Arabic attaches clitics and endings to words. Why does this give Arabic text more types? What must be the same before you compare vocabulary sizes of two languages?

Q3 More than half of the Lab 3 corpus's types occur once. Why does that make word counts *sparse*, and why is it a problem for a model that learns from counts?

Q4 When should you normalise spelling variants, and when should you use edit distance instead?

Q5 CHALLENGE In weighted edit distance, why must the first row and column add up each letter's real cost? When can swapping source and target give a different score?

Q6 CHALLENGE The table keeps only the cheapest cost in each cell. Why is that safe with the costs of this chapter? Suppose a run of insertions were cheaper, so that each insertion straight after another costs 0.5. Why would one number per cell no longer be enough, and what would each cell have to remember?

Work it through

E1 Count N and V for والطلاب يسألون عن الطلاب ("and the students ask about the students") twice: splitting on spaces only, and also splitting و off as its own token. List the types each time and give the type–token ratio. Which policy shows that the message mentions the same students twice?

E2 Fill the Levenshtein table for قلم → علم (“pen” and “knowledge”, written without vowels). Give the distance and the operations on one best path. Why does a small distance tell you nothing about meaning?

E3 **CHALLENGE** Use the written forms كاتب (*katib*, “writer”) and مكتب (*maktab*, “office”), letters in reading order. Fill the table twice: once with $c_{\text{sub}} = 1$ and once with $c_{\text{sub}} = 2$. Backtrace each one, choosing the diagonal on ties. Compare the two distances and alignments. Which one lines up the root letters ك-ت-ب?

E4 **CHALLENGE** Use these costs: delete `a` 0.5, delete `b` 2, insert `a` 1.5, insert `b` 0.25, and any substitution 3. Fill the table for `ab` → `ba`. Give the minimum cost and one edit sequence. Why does counting edits give a different answer?

E5 Add a fourth message, امتى امتحانات الترم (“the term exams, when?”), to the three messages of Example 3.2. Give *N* and *V* under P1, P2, and P3. How many times does P3 now count the stem امتحان, and why does the new message not raise that count?

E6 With the lexicon and threshold of Example 3.5, what does Sanad suggest for جدل (*gadal*, “argument”) and for غيب (*ghayb*, “the unseen”)? Are these good suggestions? Propose a rule, using the length of the query, that would make Sanad more careful with short words.

E7 CHALLENGE The reference is “I cannot open the quiz page” and the hypothesis is “I can open the quiz page now”. Align the two and compute the WER.

E8 CHALLENGE **A filter for a large lexicon.** (a) Use the lower bound of Section 3.2.2 to give the smallest possible distance between غيب and محاضره without filling a table. (b) With the lexicon and threshold of Example 3.5, which entries can Sanad skip for the query غيب without filling a table? (c) A table has $(|s| + 1)(|t| + 1)$ cells. What share of the cells does the filter save for this query? Can it ever skip the right answer?

E9 CHALLENGE **Costs learned from errors.** In 1,000 typed words whose intended letter was taa marbuta (ة), writers typed ة 880 times, ه 100 times, and ت 20 times. (a) Set the cost of typing each letter for ة to $-\ln p$. (b) Which substitution is cheaper, and by how much? (c) Insertions and deletions still cost 1. Why does that make the policy inconsistent, and how would you fix it?

Read the picture

V1 In Figure 3.7, the typo is a swap of two adjacent letters. If a swap were one operation with cost 1, what would the distance from `exma` to `exam` be?

V2 Read Figure 3.13. What does a diagonal step mean when the two letters are the same, and what does it cost?

WORKSHEET 6 / CHAPTER 4

Make the token units explicit

Chapter 4 E1 | Local objectives O1–O3



Name: _____ Date: _____

1. Apply three stated token policies

Use “Students love NLP!”. Preserve or remove spaces only as required by each policy. State whether lowercasing occurs.

Policy	Sequence / counting convention	Length
Whitespace words		
Word + punctuation		
Code points, spaces kept		

UTF-8 check: inspect `chr(0x0639)`. Record its bytes and distinguish byte count from code-point count.

2. Train five weighted BPE merges

Use the lecture corpus: low (5), lower (2), newest (6), widest (3). Start from characters plus end marker _. Count adjacent pairs using word frequencies. Break ties by first encounter in the stated corpus order.

Step	Selected pair	Weighted count	Changed segmentation / evidence
1			
2			
3			
4			
5			

3. Apply the frozen merges to lowest

Show the final pieces. Explain why application does not update pair counts and what a vocabulary needs to guarantee byte coverage.

WORKSHEET 7 / CHAPTER 4

Find the units and the sentences

Chapter 4 E3, Figures 4.4, 4.10 and 4.11 | Local objectives O2 and O5



Name: _____ Date: _____

> **Nour does not type like a textbook.** Their messages mix Arabizi, prices, links, and stretched letters, often with no full stops. Decide which units a tokenizer must keep, and where a sentence really ends.

1. Audit a space split

Message: *3ayez adfa3 150.50 gineh l-ash-shahada 15/10 #al-imtihanat 3ala www.zu.edu.eg ma sha2 Allah.* For each unit, say whether splitting on spaces *cuts* it, *glues* it to something else, or keeps it intact. Then state your policy and a task that needs it.

Unit	Cut / glue / intact	Your token policy	A task that needs it
150.50 gineh			
l-ash-shahada			
15/10			
www.zu.edu.eg			
3ayez			
ma sha2 Allah.			

2. Predict the sentence count

Use the splitter of Listing 4.1 (end mark followed by a space, or a line break). The Arabic question mark is printed here as ?. Predict before testing.

Input (transliterated)	No check	Check	True
3andi mushkila fi t-tasgil. il-mawqi3 mish biyiftah? sa3iduni!			
il-maw3id ma3a d. Ahmad is-sa3a 10.30 fi l-maktab.			
3ayiz aghayyar il-gadwal il-karneh da3 bardu			
ir-rusum 150 g.m. idfa3ha in-naharda			

3. Repair one failure without breaking another

Propose a rule that fixes the *g.m.* case and keeps *d. Ahmad* as one sentence. Give one input your rule now gets right, one it could now get wrong, and how you would record it as a test.

WORKSHEET 8 / CHAPTER 4

Round-trip and preserve meaning

Chapter 4 E2 | Local objectives O4–O5



Name: _____ Date: _____

1. Round-trip one message

Take one help-desk message. Tokenize it with a declared policy, map the tokens to IDs, and detokenize the IDs. Record everything that does not come back unchanged and the policy step that removed it.

2. Design a task-specific cleaning policy

Use the Arabic-script examples in Chapter 4. Compare stem and lemma choices for studies or meeting, stating POS/tool assumptions. Choose three lecture cleaning switches for search versus pronunciation; name a distinction that must survive.

Operation / order	Purpose / example change	Preservation check / task evidence

Checkpoint contract and untouched-source policy:

PROJECT LOG L4 / CHAPTER 4

Units of text, and a first keyboard

Lab Manual, project level L4 | Dev split, 20 tickets



Name: _____ Date: _____

> **Replace, measure, then decide.** Run `python -m sanad report` before and after each change, on the same dev split and shared parts. Keep the old version when the evidence does not support the new one.

1. Shared normaliser: v1_policies → v2_nfc

Record	Old: v1_policies	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

2. Shared tokenizer: v0_whitespace → v1_policy

Record	Old: v0_whitespace	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

3. Shared tokenizer: v0_whitespace → v2_bpe

Record	Old: v0_whitespace	New
Score(s) ± margin		
By variety		
A ticket it fixes / still fails		
Decision and why		

4. Keyboard baseline and unknown tokens

Tokenizer	Keyboard PPL (targets)	Unknown, en / msa	Unknown, egy / arabizi	Router acc.
v0_whitespace				
v1_policy				
v2_bpe				

Decision

Which versions does your `sanad.toml` select now, and which result would make you revert one of them?

REVIEW / CHAPTER 4

Tokenization, Encoding, and Preprocessing

Answer space for every item in Section 4.5, Review and practice



Name: _____ Date: _____

Quick quiz

Circle one letter for each question and give your reason in a few words.

1	A	B	C	D	Reason _____
2	A	B	C	D	Reason _____
3	A	B	C	D	Reason _____
4	A	B	C	D	Reason _____
5	A	B	C	D	Reason _____
6	A	B	C	D	Reason _____
7	A	B	C	D	Reason _____
8	A	B	C	D	Reason _____
9	A	B	C	D	Reason _____
10	A	B	C	D	Reason _____

Check your understanding

Q1 Why do words, characters, bytes, and subwords produce different lengths?

Q2 In the weighted BPE example, why does the first merge have count nine?

Q3 Can a token ID be interpreted as a semantic magnitude?

Q4 Why is stemming inappropriate as an automatic preprocessing rule for every model?

Q5 Explain how NFC, NFKC, and task-specific alef normalization differ. Why must an extracted span declare its offset unit and text version?

Q6 In the leading-boundary example, why is the third merge count three? Why does learning a piece resembling a prefix not prove a morphological analysis?

Q7 Why must a byte-level tokenizer join a token sequence's bytes before decoding them?

Q8 What does a tokenize-then-detokenize round trip test, and what does it not test?

Work it through

E1 Apply the five learned BPE merges to the unseen word lowest. Show every applicable merge and explain why the training corpus is not updated.

E2 Write separate preprocessing policies for Arabic document search and for reading that is sensitive to pronunciation. Include one transformation you would test for each.

E3 Propose one change to Listing 4.1 that fixes the جم٠ج failure without breaking the د. أحمد case. State the new rule, one input it now gets right, and one input it could now get wrong.

E4 For `"cafe\u0301 NLP"`, compute the raw and NFC code-point start indices of `NLP`. Then give the unseen `rearrange` segmentation after the four leading-boundary merges. Record which input distinctions each policy changes and one test it needs.

E5 **Count the units.** For `exam`, امتحان, and مِش (*mish*, written with one kasra), give the number of code points and of UTF-8 bytes. How many letters does a reader see in مِش? Which count would you report as "length" for a tokenizer budget, and why?

E6 **Run BPE yourself.** Learn three merges from the weighted corpus `hug`:10, `pug`:5, `pun`:12, `bun`:4, `hugs`:5, using characters, the end marker _, and the chapter's tie rule. Give each merge with its count, the final segmentation of `pun` and `hugs`, and the segmentation of the unseen word `bug`.

E7 **Predict the splitter.** Before running Listing 4.1, predict how many sentences it returns for each message, and mark any answer that is wrong:

(a) سألت أ. منى عن الموعد. لم ترد. (I asked Prof. Mona about the date. She did not reply.)
(b) “Is the portal down? I can’t log in!”
(c) “The exam is at 10.30 tomorrow.”
(d) عايز اغير الجدول الكارنيه ضاع برضه (I want to change the timetable the ID card got lost too)

E8 **Check the contract.** A model was trained with the vocabulary `[<unk>, i, love, nlp, data]` (IDs 0–4). Encode “i love nlp”. A colleague later loads the alphabetical vocabulary `[<unk>, data, i, love, nlp]`. What text do the same IDs decode to? Write one test that would catch this before deployment.

Read the picture

V1 In Figure 4.6, why are the merge counts weighted occurrences rather than numbers of distinct words?

V2 Figure 4.3 shows one text under three policies. Why can a model trained with one policy not simply be given IDs from another?

WORKSHEET 9 / CHAPTER 5

Count within the boundaries

Chapter 5 E1–E2 | Local objectives O1, O2 and O4



Name: _____ Date: _____

1. Construct unpadded and padded windows

Use “i love natural language processing”. List unpadded bigrams and trigrams. Then apply the lecture’s padding: $n-1$ start markers and one end marker for $n > 1$.

Window policy	Windows / representative boundary cases	Count
Unpadded bigrams		
Unpadded trigrams		
Padded bigrams		
Padded trigrams		

2. Reconstruct the lecture’s Sam counts

Use `<s> I am Sam </s>`, `<s> Sam I am </s>`, and `<s> I do not like green eggs and ham </s>`. Preserve casing. Explain each outgoing denominator.

Conditional	Transition count / outgoing count	Probability
$P(\texttt{I} \mid \texttt{<s>})$		
$P(\texttt{am} \mid \texttt{I})$		
$P(\texttt{Sam} \mid \texttt{am})$		

3. Compare counts with smoothing

For the workbook’s eleven-word vocabulary, calculate unsmoothed and add-one $P(\texttt{love} \mid \texttt{i})$ and $P(\texttt{data} \mid \texttt{i})$. Check the smoothed sum over all targets. Explain what a zero conditional does to a sequence probability.

REVIEW / CHAPTER 5

N-gram Language Models

Answer space for every item in Section 5.5, Review and practice



Name: _____ Date: _____

Quick quiz

Circle one letter for each question and give your reason in a few words.

1	A	B	C	D	Reason _____
2	A	B	C	D	Reason _____
3	A	B	C	D	Reason _____
4	A	B	C	D	Reason _____
5	A	B	C	D	Reason _____
6	A	B	C	D	Reason _____
7	A	B	C	D	Reason _____
8	A	B	C	D	Reason _____
9	A	B	C	D	Reason _____
10	A	B	C	D	Reason _____

Check your understanding

Q1 Why does the conditional denominator count outgoing transitions?

Q2 Why does the target support include `</s>` but not `<s>`?

Q3 In “The soups that I made from a new cookbook were delicious”, why does a bigram model have no evidence for *were* over *was*? Would a trigram model fix it?

Q4 Why can two perplexities be compared only when both models score the same targets?

Q5 What does add-one smoothing do for an unseen context?

Q6 Why can a naive fallback row sum above one? How do normalized backoff and interpolation avoid that problem?

Q7 Why should interpolation weights be chosen on development data, and why is a fluent generated sample insufficient evidence of application quality?

Q8 Why can segmenting Arabic clitics both reduce and worsen sparsity for a bigram model?

Work it through

E1 **List the windows.** Pad “i love deep learning” for a trigram model, list every prediction window, and check the count against $L + 1$. Which window lets the model predict the end of the sentence?

E2 For the two-sentence boundary example, calculate the unsmoothed probability of “i love data” including its ending. Identify one unseen transition that would make another sentence have probability zero.

E3 Using Figure 5.11, compute the unsmoothed and add-one probabilities of the second request, “I want to postpone the submission”, including its end marker. Which is larger, and why?

E4 A model assigns the targets of one held-out sentence, including </s>, the probabilities 0.5, 0.25, 0.25, and 0.125. Compute the mean negative log probability and the perplexity. What happens if one more target receives probability zero?

E5 **Score an unknown word.** Use the model of Example 5.6 ($V = 6$, add-one, <unk> in the support). Score the held-out "i love data" and the held-out "i like data", including the end marker. Give each product and each unknown-target rate. Which sentence scores higher, and why should you not conclude that the model "understands" it better?

E6 For high-order probabilities $(.75, .25, 0)$ and low-order probabilities $(.5, .25, .25)$ over A, B, C, calculate the mixture with high-order weight $.25$. Then retain discounted A/B probabilities $.6$ and $.2$, allocate residual mass to unseen C, and compare with naive fallback. Which rows can enter the chapter's perplexity calculation?

Read the picture

V1 Redraw Figure 5.5 for “i love data”. What is its probability, and what do the two sentence probabilities add up to?

V2 In Figure 5.3, why does a sentence-final appearance of a word add no outgoing event when the corpus is not padded?